

ACADEMIC SUCCESS CLASSIFIER

Predicting Student Outcomes to Improve Graduation Rates

The Challenge

- Student dropout is a major issue for universities.
- Early identification of at-risk students is difficult.

What I Built

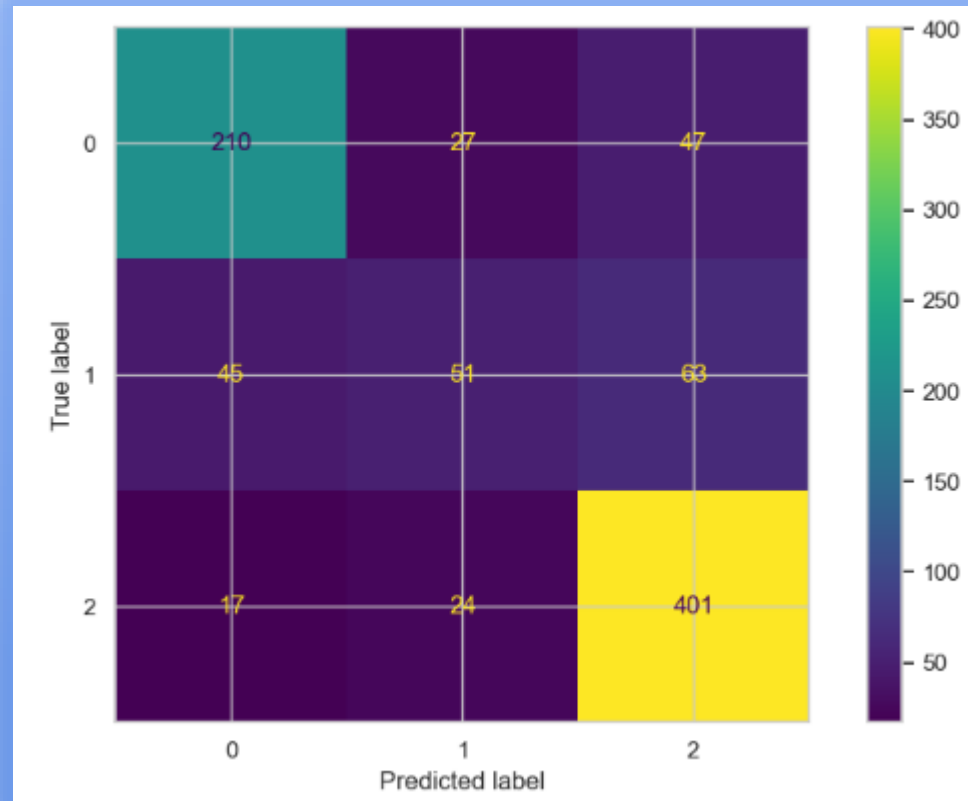
A machine learning model that predicts whether a student will:

- Graduate
- Continue
- Dropout

Results

- **~75% overall accuracy**
- Strong performance on both graduate and at-risk students

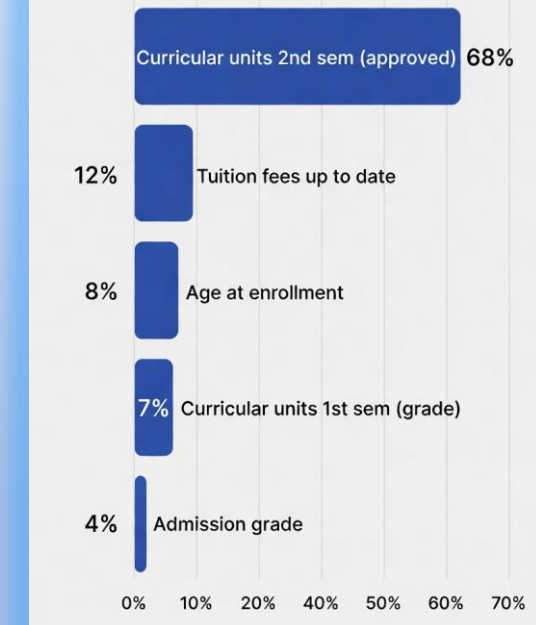
Model performance



Confusion Matrix across 3 outcome classes
(Graduate / Continue / Dropout)

Key Insight

Feature Importance



Second-semester course completion is the #1 predictor of success.